

Setting Up Constants and the Turtle Configuration

The first lines of the program cover basic setup steps for our turtle-based program:

Python

```
import turtle, math

DRAW_FRACTAL = 1 # Set to 1 through 11 and run the program.

turtle.tracer(5000, 0) # Increase the first argument to speed up the drawing.
turtle.hideturtle()
```

The program imports the `turtle` module for drawing. It also imports the `math` module for the `math.sqrt()` function, which the Sierpiński Triangle fractal will use, and the `math.cos()` and `math.sin()` functions, for the Snowflake fractal.

The `DRAW_FRACTAL` constant can be set to any integer from 1 to 9 to draw one of the nine built-in fractals the program produces. You can also set it to 10 or 11 to show the output of the square or triangle shape-drawing function, respectively.

We also call some turtle functions to prepare for drawing. The `turtle.tracer(5000, 0)` call speeds up the drawing of the fractal. The 5000 argument tells the `turtle` module to wait until 5,000 turtle drawing instructions have been processed before rendering the drawing on the screen, and the 0 argument tells it to pause for 0 milliseconds after each drawing instruction. Otherwise, the `turtle` module would render the image after each drawing instruction, which significantly slows the program if we want only the final image.

You can change this call to `turtle.tracer(1, 10)` if you want to slow the drawing and watch the lines as they're produced. This can be useful when making your own fractals to debug any problems with the drawing.